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RESEARCH ARTICLE

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A warning indicator for distribution network to extreme weather events

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Summary

Electric power distribution networks are vulnerable to natural events, especially storms and hurricanes. Because of climate change, the severity and number of natural events have been increased. Therefore, the crisis manager needs to assess and calculate the warning indicator of the system to provide proactive decisions before the event. This paper presents a model to meet this requirement. The proposed model consists of three general approaches: (a) evaluation of distribution network response to natural threats, (b) modeling the distribution network recovery, and (c) evaluation and estimation of the warning indicator level. In addition, two submodels including threat scenarios (intensity and velocity of the storm) and the fragility model of network components against threats are the inputs of the resilience assessment model. Evaluating and estimating warning indicator is carried out using damage estimation functions and a fuzzy inference system. The proposed model is applied to a real distribution network in Iran, and the results are discussed.

KEYWORDS

fragility curve, fuzzy logic, Monte Carlo simulation, resilience, warning indicator

1 | INTRODUCTION

Nowadays, the welfare, economic activities, and the proper functioning of modern societies depend on access to vital infrastructures such as electricity, water, fuel, and telecommunications. Such infrastructures are inevitably exposed to various problems and threats, including natural disasters (such as earthquake, storm, and floods), aging and degradation of equipment, consumption increment, global warming, and climate change. These challenges have increased the probability of infrastructure failures so that it is expected that the incidence and intensity of these events will be increased in the future.[1] The widespread damage to critical infrastructure can lead to extensive disasters on social and economic spheres.[2] Therefore, modeling and evaluation of the performance of critical infrastructures against all types of threats are increasingly important and need urgent planning and preparation for crisis management. Moreover, damage estimation and resilience assessment of critical infrastructures are essential to sustain operations and retrieve damaged networks after the occurrence of destructive events.[3] Electricity networks are one of the most important critical infrastructures because of the dependence of other infrastructures to electricity. Because of this, policymakers try to reduce the vulnerability of power systems to continue the operation of other infrastructures in times of disaster.[4]

The National Infrastructure Advisory Council (NIAC)[5] introduces the main features of resilience as the strength, resource adequacy, rapid recovery, and adaptability. Furthermore, in Panteli and Mancarella,[6] a conceptual structure

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is described for resilience based on the key features of resilience including strength, resource adequacy, redundancy, recovery, and compatibility. In Chaudry,[7] the U.K. Energy Research Centre (UKERC) provides the following definition for resilience: "Resilience is the capacity or capability of a system to withstand disturbances and its ability to deliver continuous energy to consumers. A resilient energy system can quickly recover from the shock and provide energy even when a disaster is happened because of an external event." The concept of resilience in power systems is characterized by a set of attributes including resistance against events, rapid recovery, compatibility, and lessons learned from occurrences in which the rapid recovery of the lost load is one of the main features of resilience.

Table 1 summarizes different aspects of resilience studies of power system related to this paper. As this table shows, the studies of power system resilience can be generally categorized into three main groups[8,15,16]: (a) hardening and prestorm preparation modeling to improve system resilience, (b) evaluating the efficiency of power system response in dealing with the threat to calculate and assess its resilience, and (c) modeling of system restoration to measure the system ability in fast recovery and normal operation condition restoration. In the first group, new studies are conducted describing utilization of distributed generation resources to strengthen network resilience.[18-20]

In the second group, using fragility models is a suitable method to evaluate system performance in response to threats. These models are based on the general concept of structural reliability.[21] Fragility curves indicate the failure probability of a component against a threat. By using these curves and Monte Carlo simulation (MCS), the system operator can estimate the damage imposed on power system.[22] The mentioned features of resilience are used in numerous studies and researches in order to assess the effects of natural threats such as earthquake, storm, and hurricane on power systems.[10,23-25] Maliszewski and Perrings[11] hypothesize that the resilience of distribution networks depends on the system infrastructure, the biophysical environment, the interactions between them, the priority of restoration, and the effectiveness of the company's response. Amirioun et al[26] propose a quantitative framework for evaluating the microgrid resilience in response to windstorms using fragility curves of distribution lines and windstorm profile.

In the third group, one of the most important issues in improving the resilience of distribution networks is the restoration of load. After natural disaster occurrence and consequently power infrastructure damage and load outages, the main task of the system operator is to recover the network in the least possible time with the aim of restoring sensitive loads and reducing economic damages imposed on the consumers and system.[8] Quiring et al[27] proposes a data mining approach, based on the classification and regression trees, to improve the predictive accuracy of the power outage models. Khomami and Sepasian[28] provide a new method for restoration and emergency reaction planning of distribution network based on the prehurricane repair team placement model, using MCS and forward dynamic programming algorithm. Furthermore, Khodaei[29] proposes a resiliency-oriented microgrid optimal scheduling model to improve the resilience of power system by curtailment reduction and local supply of loads. In line with this, a two-stage stochastic programming method is presented in Gholami et al[30] to the optimal scheduling of a resilient microgrid.

TABLE 1 Classification of power system resilience studies

Categories of Power System Resilience Studies[8]	Short-Term Resilience Measures after the Weather event[9]	Key Factors for Assessment of System Resilience	Models of Power System Damage, Outage Duration, and Restoration[8]	Power System Restoration Techniques[8]
• Hardening and pre-storm preparation models • Damage assessment • Corrective actions, emergency response, and system restoration	• Disaster assessment and priority setting • Restoration of damaged components, such as poles and conductors • Traditional restoration actions, eg, reenergizing transmission and distribution line, unit restarting, load restoration, and resynchronization of areas	• Economic losses[10] • Outage duration[11] • Expected annual resilience[12] • Resilience achievement worth index[13] • Loss of load expectation[14] • Expected energy not supplied[15] • Performance index[16] • Resilience triangle[17]	• Statistical models including data and parameters, data fitting models, and measurement of fitting goodness and example • Simulation-based models	• Conventional restoration strategies such as expert systems, heuristic approaches, and mathematical programming • Distributed generations (DGs) and microgrids for load restoration • Advanced distribution automation techniques and decentralized restoration strategies

One of the most important challenges of distribution companies is the prediction of the severity of damage before the occurrence of a destructive event.[6] In this regard, some researches propose warning indicators.[29,31] However, the development of such indicators for distribution network operation is not widely considered in the literature. Therefore, in this paper, a method is presented that can be used to warn the system operator about the occurrence of a natural disaster and network destruction.

This paper presents a comprehensive and multistage model for risk assessment and calculating warning indicators. This model includes the evaluation of system response to the threat, modeling of system restoration, and calculation of warning indicators where fuzzy logic and damage estimation functions are used. Moreover, in order to evaluate the damage imposed on the network, the MCS and fragility curves are used. By modeling the behavior of crisis managers, an innovative algorithm is designed to simulate the system recovery process.

Considering the literature review and Table 1, the contributions of this paper can be expressed as follows:

1. There are different restoration techniques. This paper introduces a new prioritizing method that arranges the equipment repair priorities based on a new index.
2. In this paper, restoration time and restoration cost are considered as key factors for assessment of system resilience. Furthermore, the damage estimation function is introduced, which is obtained using curve fitting techniques. Application of this function considerably reduces the computational burden of damage estimation.
3. While various researches are conducted on the topic of system resilience, there are hardly any warning indicator for assessment of distribution network resiliency. Therefore, another achievement of this paper is the presentation of a method based on the fuzzy inference system for warning notification before the damage caused by a natural disaster.
4. While most of the related papers studied the resilience of power system, the proposed scheme of this paper is applied to a real distribution network in Iran.

Section 2 of the paper introduces the proposed model. Section 3 presents and discusses the simulation results, and Section 4 presents the conclusions obtained from the paper.

2 | PROPOSED MODEL

From the viewpoint of resilience studies, the exposure of distribution networks to destructive events such as storm and hurricane can be classified into three stages, namely, before the incident, failure expansion, and restoration.[32] These steps explain the main characteristics of system resilience including resistance, redundancy, and response/recovery.

The proposed model consists of three general approaches: (a) evaluation of distribution network response to natural threats, (b) modeling the distribution network recovery, and (c) evaluation and estimation of the warning indicator level. Moreover, two submodels including threat scenarios and the fragility model of network components against threats are the inputs of the resilience assessment model as shown in Figure 1. Each step of the model is described as follows:

In the first step of the proposed model, in order to evaluate the system resiliency against the upcoming threat and determination of damaged components and lost load, the fragility curves of the poles and conductors are extracted using MCS. In the second step, an innovative approach is provided to prioritize the recovery process of damaged feeders using resilience criterion. In the third step, the proposed model evaluates the warning indicator and estimation functions using the Mamdani fuzzy model.

2.1 | Threat specifications

The assessment of the resistance and resilience of the distribution system against threats with different severity and estimation of the required time for distribution network restoration are two of the objectives of the model presented in this paper. In general, there are two approaches to model the threat of storm and hurricane. The first one is to adopt deterministic scenarios that model a set of historical incidents, and the other is to outline probabilistic scenarios that predict how future threats will occur.[15]

In this paper, with the aim of evaluating the maximum potential of resilience and adaptability of a distribution network, incremental coefficients are applied to the values of the historical data (storm speed) to obtain new deterministic

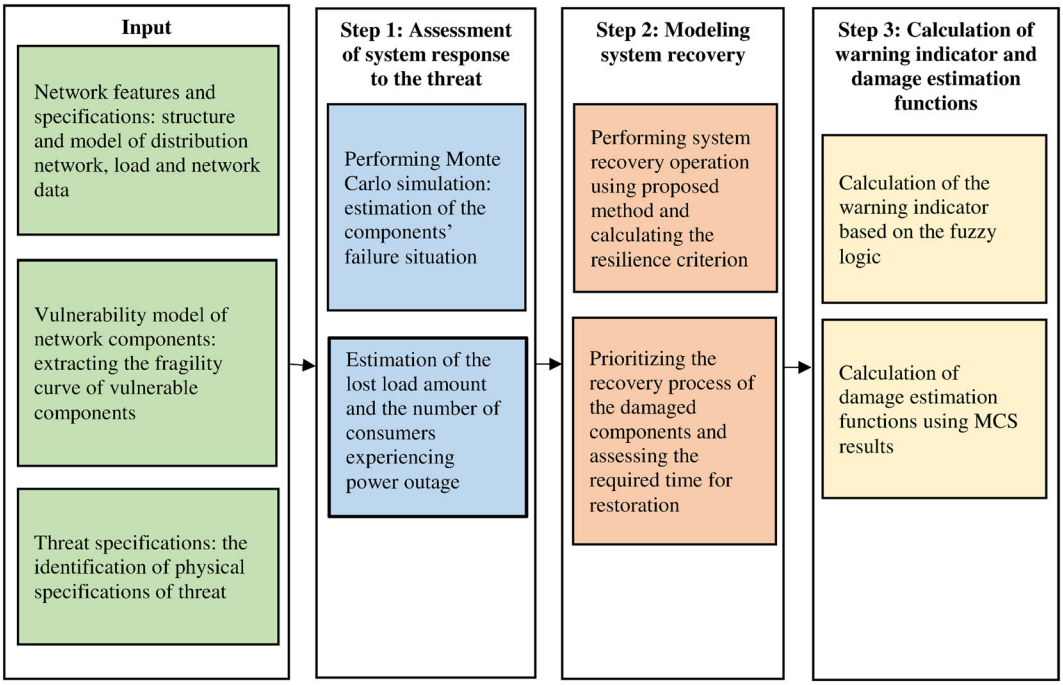


FIGURE 1 The structure of the proposed model

values as the input of the model. Therefore, in the simulation, the wind speeds are considered as in Table 2. As this table shows, Mazandaran province, the studied region in this paper, is a windy area with wind speeds of more than 40 m/s.[33]

2.2 | Fragility curve

The fragility curves indicate the possibility of a component failure against an event occurrence.[9,14] These models are derived from the concept of structural reliability.[21] In the present paper, the fragility analysis method is used to calculate the probability of failure of the poles and the conductors to the storm. In order to extract the fragility curves, it is necessary to first obtain the mathematical models of these curves. These models can be extracted by fitting historical and experimental data of conductors and poles of medium voltage network to a curve. To obtain the fragility curve of each component from the historical data, the failure percentage of poles and conductors are mapped to the wind speeds because for each wind speed, there is a related failure rate. After that, using a curve fitting method, the most appropriate curve is estimated for each component. It is observed that the fragility curves obtained for medium voltage conductors and poles of the test system follow exponential functions.

The fragility curve used in this paper is an exponential function (see Equation 1), which is obtained from the studies carried out by Mazandaran Power Distribution Company.

$$Fc_q = A_q.\mathrm{e}^{B_q.Ws}, \tag{1}$$

where A_q and B_q are usually obtained experimentally by examining the historical data.

TABLE 2 Applying threat incremental coefficients to wind speed

Normal Conditions	×1.25	×1.5	×1.75
40 m/s	50 m/s	60 m/s	70 m/s

2.3 | Assessment of system response to the threat

By performing MCS and using storm specifications and fragility curves of poles and conductors, the failure status of each component is determined in each iteration. Thus, with regard to the failure situations of components, the system operator can estimate the amount of lost load, the number of out-of-service consumers, and the values of system indices like energy not supplied (ENS) and the number of out-of-service substations and transformers.

2.4 | System restoration modeling

Restoration time is greatly dependent on the extent of the failures, the number of human resources, availability of spare equipment, and the adequacy of transportation systems/infrastructure. Furthermore, emergency response, as an effective strategy for recovering the system from crisis situations, should be applied as quickly as possible. For that reason, this paper provides a network restoration algorithm that facilitates system recovery in the shortest possible time. In the proposed algorithm, the most important issue is to prioritize the repairs of damaged components. This prioritization should be planned considering network structure and identification of critical and sensitive points.

Each distribution network typically consists of several feeders and subfeeders where each subfeeder consists of a number of lines and poles. In recovery planning and implementing of a damaged network after a catastrophe, the direction of network repairing is from the substation to distribution nodes. However, the recovery priority of each node can be determined using various indices considering technical, organizational, social, and economic characteristics. In Equation 2, a social-technical prioritizing criterion is proposed that can maximize the number of repaired distribution transformers in the minimum possible time. In fact, using this index, restoration of technical equipment and consumers are simultaneously considered.

$$\mathrm{Pr}_l = \frac{\dfrac{Nt_l}{\sum\limits_{q_l \in Q_l} Rt_{q_l}}}{\sum\limits_{l \in L} \dfrac{Nt_l}{\sum\limits_{q_l \in Q_l} Rt_{q_l}}}, \tag{2}$$

where the feeder with higher value of index has higher priority. A similar index can be also defined for prioritizing subfeeders. The required time to restore each pole or conductor follows a normal probability distribution function (PDF) as in Equation 3.[34] Using this equation, the required time for fixing the failure of each damaged pole or conductor is obtained using random number generation[35] from the normal PDF in each MCS trial.

$$Rt_q = f\left(x|\mu_q, \sigma_q\right) = \frac{1}{\sigma_q\sqrt{2\pi}} e^{\frac{-(x-\mu_q)^2}{2\sigma_q^2}}, \tag{3}$$

where the values of μ and σ can be obtained from the information provided by the distribution company.The total cost of the recovery can be calculated from Equation 4:

$$Tc = Nu.Cu + \sum_j Ep.En_j + \sum_j Ve.En_j + Vs. \tag{4}$$

In Figure 2, the proposed flowchart for network recovery problem is shown. The flowchart includes three main levels for repairing based on Equation 2: (a) prioritization of feeders, (b) prioritization of subfeeders of each feeder, and (c) prioritization of components of each subfeeder. Therefore, the recovery process commences from the feeder with the highest priority. In this feeder, the subfeeder with the highest priority will be selected. Then, the repairing process will be started with the most important poles and conductors. The process will continue for other subfeeders and feeders.

2.5 | Calculation of warning indicator and damage estimation functions

In this paper, using the fuzzy logic and based on expert judgments, distribution company's business strategies, and the severity of the crisis, the warning levels are defined as in Table 3. The levels of warning indicator can be defined

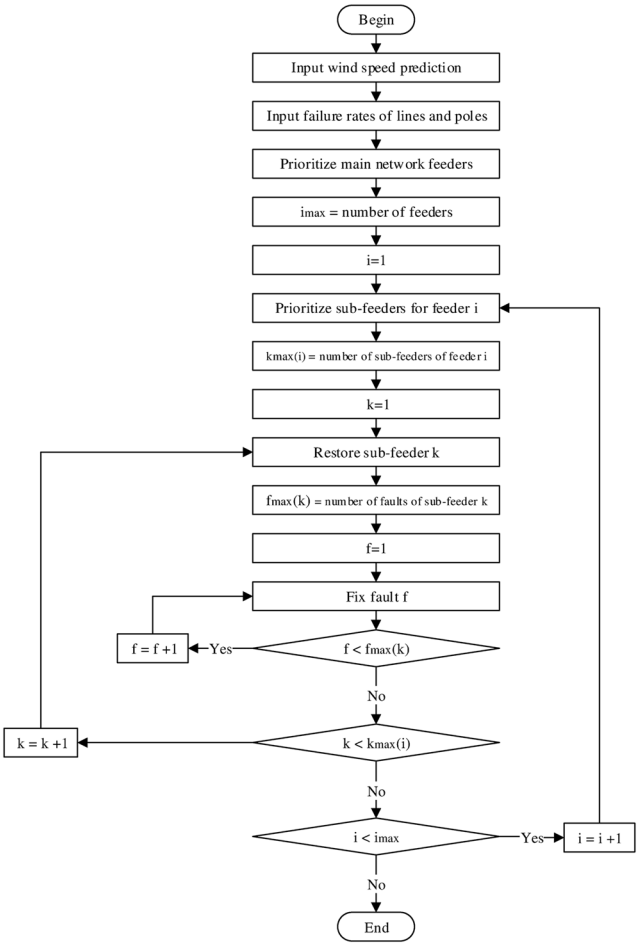


FIGURE 2 The proposed algorithm for network recovery

TABLE 3 Qualitative levels of criticality of condition

Recovery Time	Recovery Cost			
	Low	**Medium**	**High**	**Very High**
Low	Alarm	Alarm	Semicritical	Critical
Medium	Alarm	Semicritical	Critical	Super-critical
High	Semicritical	Critical	Super-critical	Ultra super-critical
Very high	Critical	Super-critical	Ultra super-critical	Ultra super-critical

differently in various regions because of the different technical, social, and economic indices. In this paper, fuzzy membership sets are defined for recovery cost and recovery time, as shown in Figures 3 and 4, respectively. It should be noted that the threshold values for recovery costs are determined based on the budget of emergency conditions and crisis management where the value of recovery cost is supposed to be between zero and one that is related to the company's allocated budget for the crisis. Furthermore, the criticality index is defined as a number between zero and one, which indicates the severity of the network's criticality (see Figure 5). As shown in Table 3, the crisis situation and the warning conditions are ranked by defining different qualitative levels for two characteristics, ie, recovery cost and recovery time. Using this table, the state of the upcoming crisis can be predicted for different wind speeds. In fact, Table 3 describes the rules associated with the proposed fuzzy model. In a fuzzy system, rules can be applied using different logical operators, such as AND and OR. In this paper, the Mamdani fuzzy model is considered by using AND operator to combine inputs. Figure 5 shows the output membership functions (criticality index).

The proposed algorithm for predicting the criticality of the network condition against natural disaster is shown in Figure 6. Indeed, based on Table 3, the model presented in Figure 1, and damage estimation function, the criticality

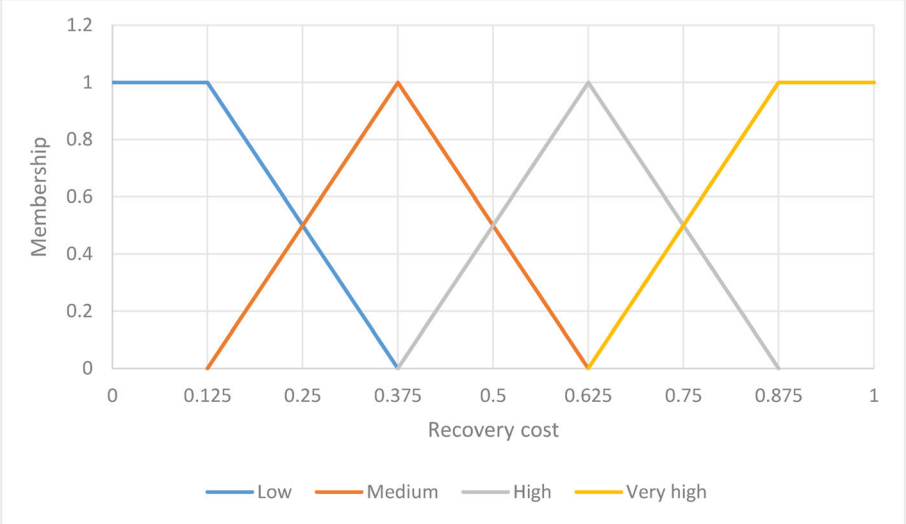


FIGURE 3 Membership functions related to the recovery cost

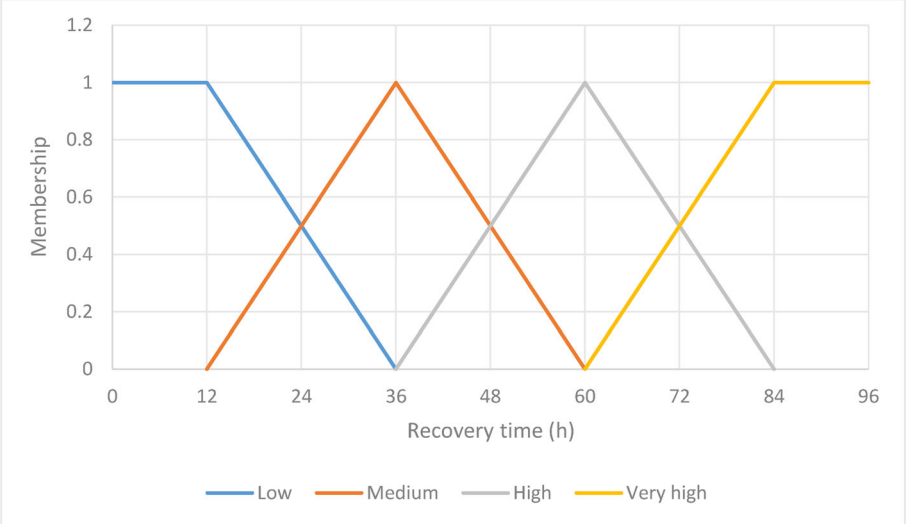


FIGURE 4 Membership functions related to the recovery time

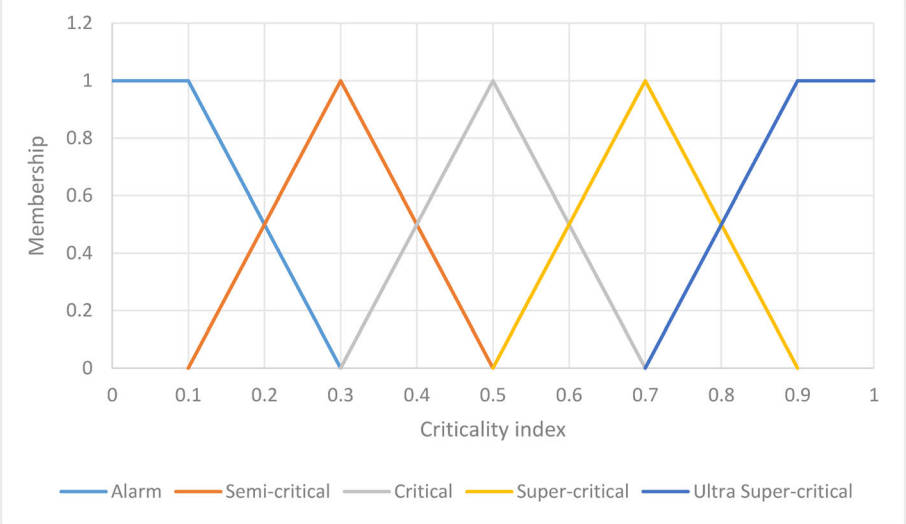


FIGURE 5 Output membership functions related to the criticality index

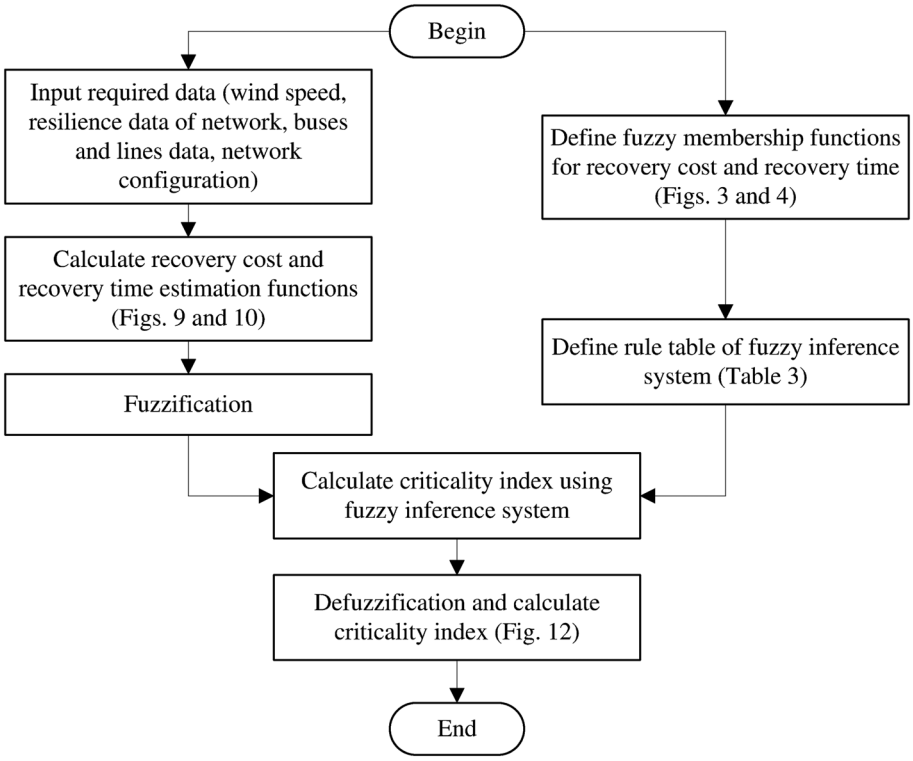


FIGURE 6 The proposed algorithm for estimation of criticality index

of the situation can be evaluated at different speeds of the storm. Thereby, the required preventive actions can be performed. It should be noted that the damage estimation function is obtained from MCS results using curve fitting techniques (see Section 3).

3 | NUMERICAL RESULTS AND DISCUSSION

A test system with 10 medium voltage feeders and two 63/20-kV substations in Mazandaran province in Iran is selected for the case study. Table 4 presents system data, and Figure 7 illustrates the schematic diagram of network.

3.1 | System response assessment

The test system is located in a small town with rural areas, many agricultural loads, and considerable population density. In this paper, the fragility curves of poles and conductors are considered as in Equation 1, which are inquired from the relevant distribution company. Table 5 demonstrates the parameters associated with these curves.

The first step of the proposed model is the assessment of the system response to threat (see Figure 1). In this step, the assessment of the medium voltage network vulnerability is done using the fragility function and MCS. In each iteration of MCS, the situation of failure of conductors, poles, and transformers is obtained. Thereby, total lost load and number of out-of-service consumers are determined.

TABLE 4 Specifications of test system

Total Load Demand of Feeders	Number of Distribution Transformers	Number of Poles	Network Length	Number of Feeders
110 152 kW	486	1600	80 km	10

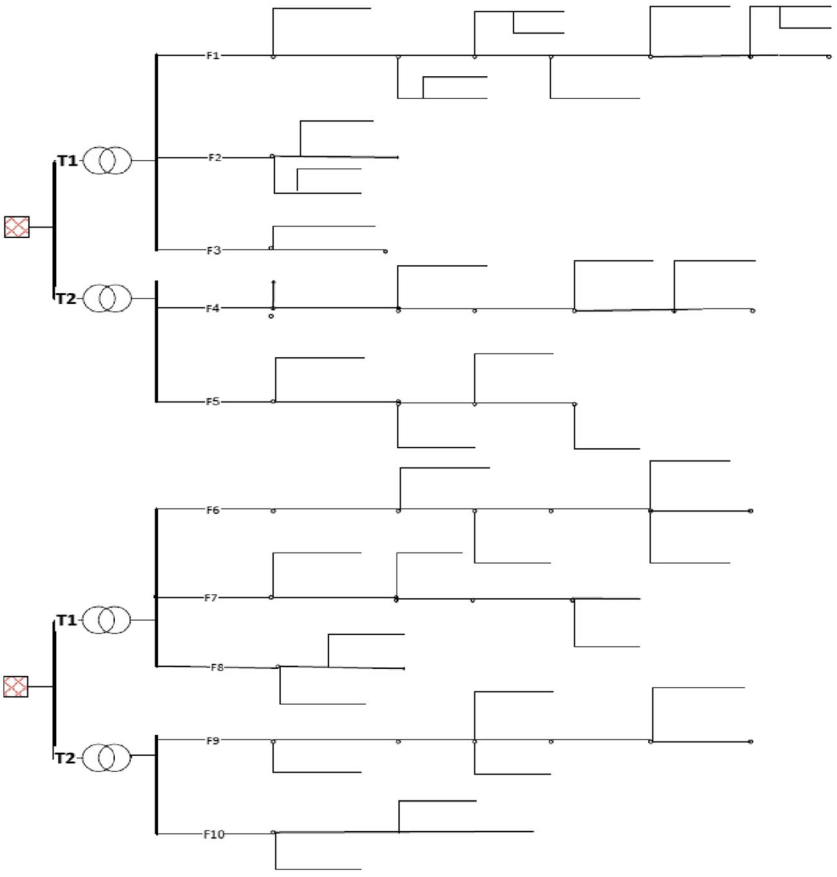


FIGURE 7 Schematic diagram of test system

TABLE 5 Parameters related to the fragility curves of poles and conductors

Pole failure	A_q	0.004768
	B_q	0.07249
Conductor failure	A_q	0.006727
	B_q	0.1114

3.2 | System recovery modeling

Recovery modeling is the second step of the proposed model. In this step, prioritizing the recovery of feeders and subfeeders and system recovery operations will be carried out as shown in Figure 2. In recovery modeling, according to the actual situation in the relevant area, improvement of emergency response is obtained by increasing the number of operational units in the recovery period (see Table 6). Table 7 presents the estimated required time and cost for fixing the failure of each pole and each conductor. The problem is solved for 5000 trials of MCS where the convergence condition is obtained. The obtained results of costs and times for restoring the network for different wind speeds are presented in Table 8.

TABLE 6 Number of operational units after event occurrence

Time After Event Occurrence	Number of Operational Units
First day	3
Second day	6
Third day	12
Until the end of the crisis	12

TABLE 7 Repair times and costs for restoring each pole and each conductor

Equipment	Repair Time	Repair Cost
A medium voltage pole	$Rt_{\text{pole}} = f(x\vert\, 4\text{h}, 2\text{h})$	1000$
A section of medium voltage overhead line	$Rt_{\text{conductor}} = f(x\vert\, 2\text{h}, 1\text{h})$	250$

TABLE 8 Recovery costs and times for different wind speeds

Wind Speed, m/s	40	50	60	70
Recovery time, h	45	65	94	132
Recovery cost, $	56 212	83 332	120 770	177 070

Figure 8 depicts the curves of load restoration during the recovery time for different wind speeds. This figure demonstrates how much ENS is recovered at different times after the storm. Finally, this figure is used to compute the warning indicator of the critical situation presented in step three of the model. This figure shows that with increasing the wind speed, the recovery process takes longer time and slope of curves changes. At any given time, with dividing the amount of unsupplied load into the total load of network, the percentage of unsupplied load can be calculated.

3.3 | Calculation of warning indicator

In this subsection, the calculations of warning index based on MCS results, recovery cost, and recovery time are discussed. Extracting the damage estimation functions and calculating the warning index and the critical state declaration is the third step of the model presented in Figure 1. For calculating the cost and time estimation functions using curve fitting methods, the data of Table 8 are used. It is observed that both estimation functions follow the exponential function. Table 9 presents the parameters of estimated exponential functions of recovery cost and recovery time. In order to assess the suitability of fitness, goodness of fit criterion is tested on the extracted functions. One of the advantages of extracting damage estimation functions of cost and time is that the system operator does not need to run an MCS any time to evaluate the models of vulnerability and recovery.

According to Table 3, two factors, ie, recovery time and recovery cost, are used to predict the extent of the crisis. Figure 9 shows the variation of recovery cost of distribution network for different wind speeds. Figure 10 also demonstrates the recovery time of network at various wind speeds, where the exponential function of the estimated recovery time is calculated in terms of speed. Moreover, Figure 11 shows that the recovery cost in terms of recovery time is also an exponential function where with increasing recovery time, the recovery cost is exponentially increased.

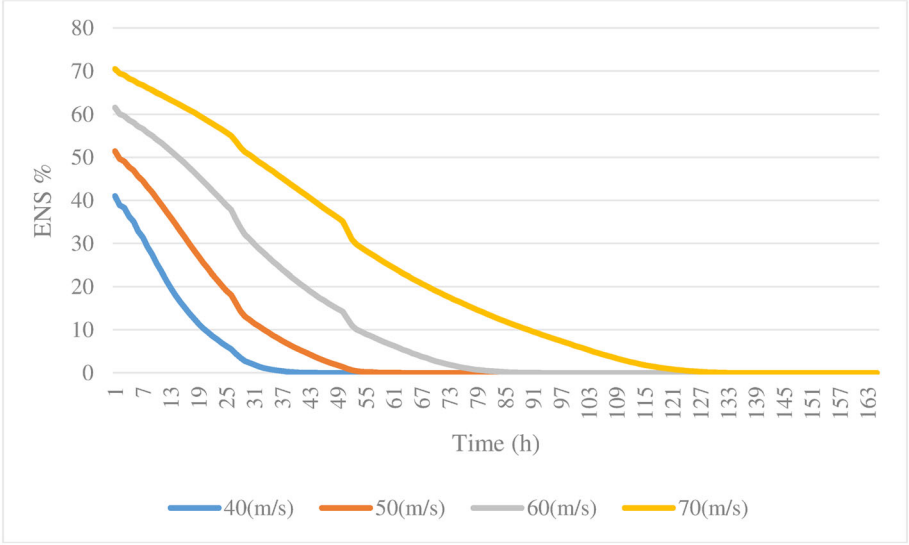


FIGURE 8 Energy not supplied during the recovery time

TABLE 9 Parameters of exponential estimation functions of recovery cost and recovery time

Recovery time	A	31.5778
	B	0.3597
Goodness of fit criteria	R_{square}	0.9996
Recovery cost	A	38.562
	B	0.3813
Goodness of fit criteria	R_{square}	0.9999

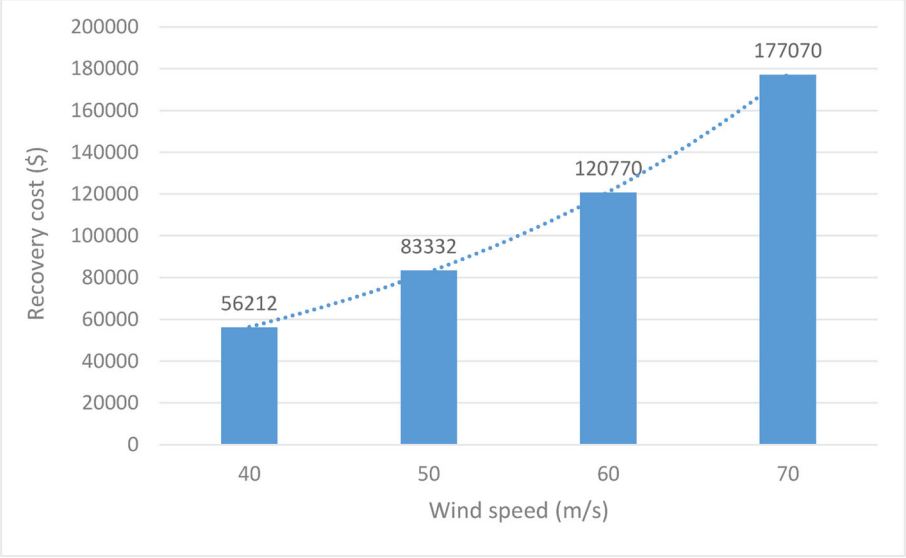


FIGURE 9 Comparison of recovery costs at different wind speeds

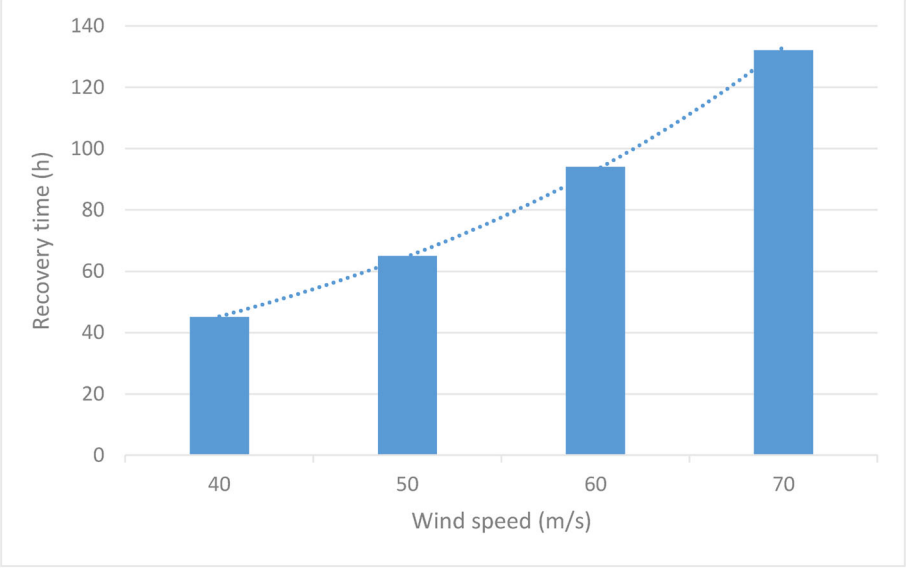


FIGURE 10 Comparison of recovery times at different wind speeds

Calculating the warning indicators and reporting the crisis situation is another part of the third step of the model, which is based on the results of the simulation and according to the criteria listed in Table 3. According to Figure 11 and Table 3, the criticality of the situation can be estimated as in Table 10. Therefore, the criticality of the conditions is considered to be proportional to the variations of wind speed and predictive judgments of the network situation.

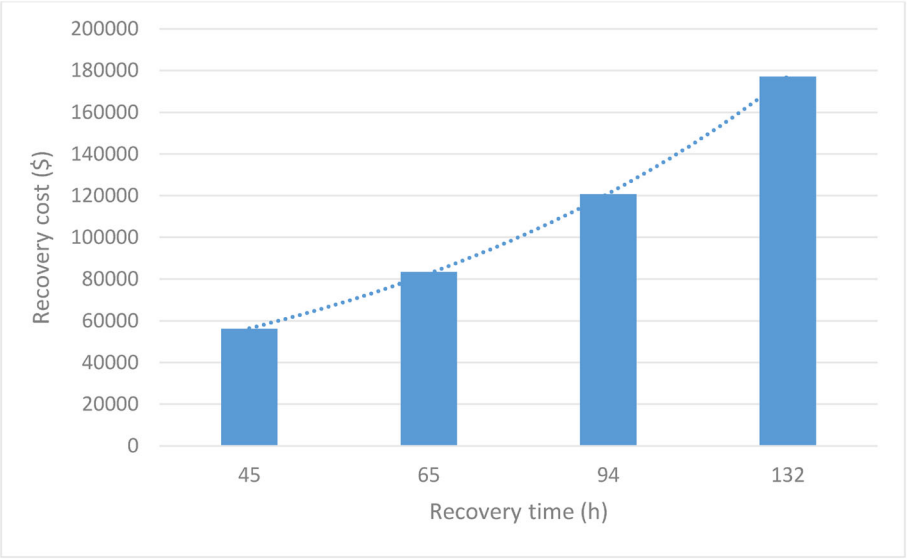


FIGURE 11 Recovery cost against recovery time curve

TABLE 10 Network status according to the crisis index

Crisis Index	Network Status
Less than 0.2	Alarm
0.2 to 0.4	Semicritical
0.4 to 0.6	Critical
0.6 to 0.8	Super-critical
More than 0.8	Ultra super-critical

The output of the defined fuzzy system is shown in Figure 12. This figure is obtained using Table 3 and Figures 3, 4, and 5, where for each recovery cost and recovery time (obtained to wind speed), a number between zero and one is attributed to the output, which indicates the criticality of the situation. With increasing each parameter of recovery cost or recovery time, the network status will be more critical.

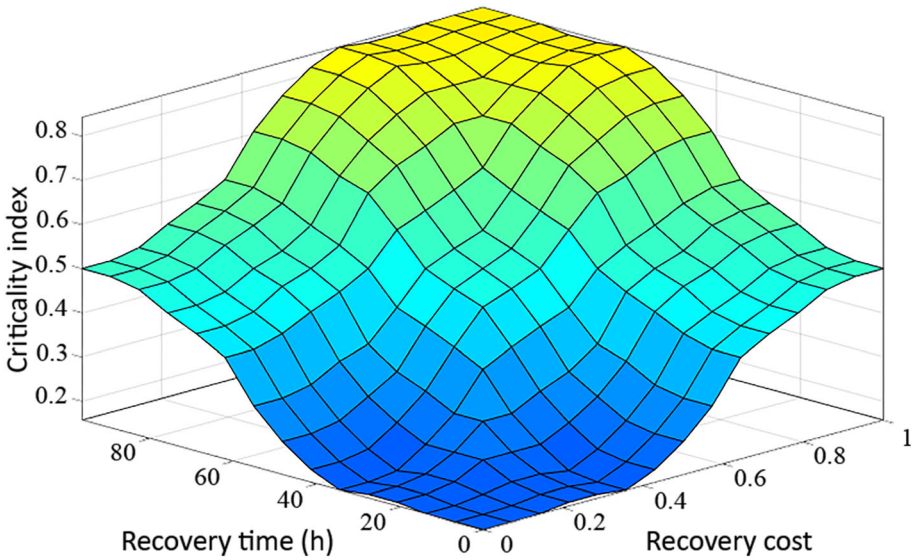


FIGURE 12 The levels of network criticality

4 | CONCLUSION

In this paper, a three-level model was introduced that presented predictive declaration of the warning level of distribution networks exposed to the hurricane. This model can improve the resilience of system and reduce the recovery time of consumers. Also, a fuzzy logic was used to calculate the warning indicator of system. The main advantages of the suggested model are as follows:

- Previous studies in this area have mainly focused on the reliability and risk studies, while the main focus of the model presented in this paper lays within the scope of distribution network resilience.
- The proposed method for modeling the recovery of distribution networks is completely operational because it is straightforward and practical. Therefore, it can be implemented in less developed countries to predict the criticality of network.
- The suggested model presents an appropriate repair prioritization method that leads to recover the maximum number of transformers in the shortest possible time. As a result, using this method, the technical, economic, and social characteristics of network can be considered for restoration.
- Using curve fitting method and estimation of damage functions and required time for recovery are innovative aspects of model that can effectively reduce computational burden of problem.
- If wind speeds are suitably predicted, the system operator can manage the crisis.

The topic of the future work of the authors is to present a bi-level optimization-based model for reconfiguration of the distribution network to improve the system resilience against severe weather events with the aim of minimizing the cost of load outage.

LIST OF SYMBOLS AND ABBREVIATIONS

Indeces

J index for load.
L index for distribution feeder.
Q index for equipment.

Parameters

A and B parameters of exponential function.
Cu cost of each service unit including car and equipment ($).
Ep energy price ($/kWh).
Nt number of transformers.
Nu number of service units.
Ve value of energy not supplied from viewpoint of consumers ($/kWh).
Ws wind speed (m/s).
μ and σ mean value and standard deviation associated with repair time of defective components (h).

Sets

L set of network feeders.
Q set of equipment, ie, pole and conductor.

Variables

En energy not supplied (kWh).
Fc function of fragility curve.

Pr prioritizing criterion of repairing.
Rt required time for recovery (h).
Tc total cost of recovery ($).
Vs value of substituted equipment ($).

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